Sequential Circuits: Latches

In this lab, you've learned about edge sensitive circuits and explored some of the power therein.

Rubric

Item	Description	Value
Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
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Names

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Summary

In this lab, we used Vivado to design and test 3 types of flip-flops: D, JK, and T, on the Basys3 board. We used switches for inputs, buttons for clock signals, and LEDs to display the outputs. This lab helped us better understand how flip-flops store data, respond to clock signals, and work in sequential logic circuits.

Lab Questions

What is the difference between edge and level-sensitive circuits?

An edge-sensitive circuit only changes the data when the clock switches from high or low. A level sensitive circuit works with looking at what the input is when it is in between each clock signal.

Why is it important to declare the initial state?

It needs an initial state to know where to start when it is powering on for example without this it wouldn't know where to start.

What do edge sensitive circuits let us build?

An edge sensitive circuit allows us to build flip flops. By limiting the responsiveness of a latch to the moment a clock signal switches from high to low or low to high.